

Chapter 2, Problem 2.1E

Explanation of Solution

A system to control anti-lock braking in a car:

- The most appropriate generic software process model for controlling anti-lock braking in a car is Waterfall model.
- The anti-lock braking is a safety-critical system.
- In the waterfall model, for the proper functioning of the anti-lock braking system of the car, the analysis and design requirements must be scheduled without any flaws before the system implementation.

A virtual reality system to support software maintenance:

(C) Travel planning system–

- This system will assist the users to plan their travel program with the lowest environmental impact and cost-efficient manner.
- This system will depend on the user's interaction and input which means it will require enormous testing throughout the development process to ensure the performance and efficiency of the system in different environments and as per the user's needs.
- So this system will comprise complex UI but must be reliable and stable.
- Therefore, an incremental development model will be suitable for the development of this system because the requirements of the system will change after getting real user experience.

Scenario#1

You are the owner of a software engineering company. Your employees want you to pay for them to attend training session .How would you respond in a way that is legal, moral and ethical?

- Due to company is running in loss from few months company. Company is unable to approve your request in positive way. Even then they want to attend they can attend by own company encourage them.
- When the company will be in good condition they can attend and company will be pay for them.

Scenario#2

You are the owner of a software engineering company. Your employees want you to let them do pro bono work for a local non-profit organization on company time. How would you respond in a way that is legal, moral and ethical?

- Its a good thing that your employees are willing to do pro-bono work. As team constantly over-delivers and can keep that going even while doing the pro bono work, there is no blockage here. Team will learn from that job that will benefit their professional job. There might be advantages in having them experience a different environment.
- We can use this pro bono work to generate free publicity

Scenario#3

You are a software engineer at a company where management routinely encourages you and your colleagues to use pirated software. How would you respond in a way that is legal, moral and ethical?

- That's highly illegal. You should notify them that it is illegal.
- It will be very expensive and damaging to company reputation. Pirated software have different issues like they are:
 - Illegal
 - Risky
 - Have less security
 - Easily hack
 - Unproductive

Question #3

You have been appointed a project manager for a major software products company. Your job is to manage the development of the next generation version of its widely used word-processing software. Because competition is intense, tight deadlines have been established and announced. What software process model(s) would you choose and why?

Assumptions

We have to use that process model which meets the requirements to make the version in the given time limit because the deadline has been defined already.

Selected Process Model

We will use the RAD (Rapid Application Development) model because tight deadline has been announced for project completion. We need to modify the previous version of word-processing software. As Rapid Application Development is a linear sequential software development process model that emphasises an extremely short development cycle.

A component-based construction approach is used. There can be several teams working on different components during this time limit announced.

Reason for selection:

The main reason for choosing

Question# 4

You have been asked to develop a small application that analyzes each course offered by a university and reports the average grade obtained in the course (for a given term). What software model(s) would you choose and why?

As the requirements are simply known and understood that we have to just develop a small application that analyzes each course offered by a university and reports the average grade obtained in the course

Selected Process Model:

The sequential/Waterfall Model is better for this application. Because water fall model is used when the requirements are well understood for the start and no changes are needed in that so as the requirements of the application is well understood for the start so waterfall model is suitable for this.

System need modification and integration .System is large and use in public sector .University of Gujrat all department use this system according to their needs, which mean departments have some same function and some different function. Risks analysis is also important factor which should be resolve in given time frame. As system use in public sector, so verification and validation factors must be administered accordingly.

Selected Process Model:

Selected process model is "Component Model" use to develop this system.

In the majority of software projects, there is some software reuse. This often happens informally when people working on the project know of designs or code that are similar to what is required. They look for these, modify them as needed, and incorporate them into their system. A general process model for reuse-based development. Although the initial requirements specification stage and the validation stage are comparable with other software processes, the intermediate stages in a reuse oriented process are different. These stages are:

1. Component analysis given the requirements specification, a search is made for components to implement that specification. Usually, there is no exact match and the components that may be used only provide some of the functionality required.
2. Requirements modification during this stage, the requirements are analyzed using information about the components that have been discovered. They are then modified to reflect the available components. Where modifications are impossible, the component analysis activity may be re-entered to search for alternative solutions.
3. System design with reuse during this phase, the framework of the system is designed or an existing framework is reused. The designers take into account the components that are reused and organize the framework to cater for this. Some new software may have to be designed if reusable components are not available.
4. Development and integration Software that cannot be externally procured is developed, and the components and COTS systems are integrated to create the new system. System integration, in this model, may be part of the development process rather than a separate activity.

Component software engineering has the obvious advantage of reducing the amount of software to be developed and so reducing cost and risks. It usually also leads to faster delivery of the software. System boundaries should be defined and clearly specified in a component-based approach requirement analysis also implies that it is necessary to analyze whether requirements can be fulfilled with available components. System is use in public sector. All departments have some same function and some different function .We use component base approach to reuse the same component in this way we reduce our code and save time. This becomes important as we complete project in this year. The risk is analysis in every component by doing this risk is reducing and project become more accurate and efficient. This project approach is as useful when we need to modifier the some component we do not need to modify all project we just modify that component. Another reason for using this approach is to save time and complete this project in given time frame, risk is also reduce by this approach. Validation and verification also made on each component.

Question #1

You have been appointed a project manager within an information systems organization. Your job is to build an application that is quite similar to others your team has built, although this one is larger and more complex. Requirements have been thoroughly documented by the customer. What software process model(s) would you choose and why? Explain in detail.

Answer:

As the information system already exists in which some development is to be made and released. Now Developer just need to make some addition in the previous application because Customer requirements are documented, which mean requirements are well understood and have no need to modified requirement.

Process Selected:

Prototype model will be used in development of this application. In prototype model application or project is made in series of increment throughout project. In prototype model we first make a prototype and give to the customer and make changes in it accordingly as required by the customer. Further increment made after previous ones. Large systems are particularly suitable for prototype model.

Prototype model is a strategy that allows system to be developed in pieces. Prototype model allows the additions in process as per requirements, process change can be implemented.

To make this application we have to use the already existing application and make increments in it and made new more complex and large application Requirements are fulfilled already for this application.

Question# 2

University of Gujrat is currently running through different departments like Administration, Accounts, Examination, Admission, Library, Computer Labs, Faculty Management, and Student Management etc. Every department has its own specific processes and each department is using computer-based system to some extent but not complete computer based solutions. There is also inter-departmental communication for the smooth running of all functions in respective departments. It is decided by the higher management that all the departments should be integrated under one system and that system should accommodate all the processes existing in all departments. It is also decided that if this system is implemented successfully then it will also be used in other departments of University of Gujrat independently after configuring it according to their needs. Higher Management wants to see this system within this year. Risks, which can arise, should be accommodated implicitly keeping the time factor in mind. As the system will be used in public sector, so verification and validation factors must be administered accordingly. What software process model(s) would you choose and why? Explain in detail

Scenario #1

- ▶ **You are the owner of a software engineering company. Your employees (engineers) want you to pay for them to attend training.**
- ▶ **How would you respond in a way that is legal, moral, and ethical?**

If you need to take part, you will have to pay for a legal, moral and ethical presence. If there is something that they do not need to participate, and they are doing their own, you are not able to pay them, however, if you want to do so, it will definitely encourage. , Or if you refuse it, it will surely reduce the motivation.

Scenario #2

- ▶ **You are the owner of a software engineering company. Your employees (engineers) want you to let them do pro bono work for a local non-profit organization on company time.**
- ▶ **How would you respond in a way that is legal, moral, and ethical?**
It is not legal, ethical or moral judgment - it is a business decision. You are giving them time to work effectively; therefore, in fact, make a charitable contribution to your organization that they are working. It's like donating - you should decide that you can donate it. If they wanted to do this on their own time then it would be a different story.

Scenario #3

- ▶ **You are a software engineer at a company where management routinely encourages you and your colleagues to use pirated software.**
- ▶ **How would you respond in a way that is legal, moral, and ethical?**

Tell your management that if the company is using pirated software for free, it would be only honest to freely distribute all the software produced by the company. Of course, it still wouldn't be ethical, the same way it is not ethical to give away the items you steal in a shop, but at least it would be equal.

(B) VR(Virtual Reality) system–

A VR(virtual reality) system for supporting the software maintenance process will be dynamic and comprise extensive UI components.

- A VR(virtual reality) is itself a new and separate piece of software that will comprise many different dynamic components.
- So incremental development approach with UI prototyping will be an appropriate model for the development of this virtual reality system.
- Hence, the Agile model that comprises an incremental approach with UI prototyping will be the best model for managing the development process of this system

An interactive travel planning system that helps users plan journeys with the lowest environmental impact:

- The most appropriate generic software process model for an interactive travel planning system is the prototype model and incremental development model.
- The requirements of the user may change and fast delivery is essential to be implemented.
- Therefore, the prototype model will save the development time to focus on essential function first to be later completed.

A University accounting system that replaces an existing system:

- The most appropriate generic software process model for a University accounting system that replaces an existing system is Waterfall model.
- Because of the existing system, the system requirements can be predicted.
- This is mission-critical system and the requirements are stable and reusable.

A virtual reality system to support software maintenance:

- The most appropriate generic software process model for a virtual reality system to support software maintenance is Incremental development.
- The system requirements keeps on changing and it cannot be predicted before the implementation and it requires complex programming for the software.
- This software is not a critical system.